Predict a position and a range

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Learning target: Predict a future position and explain the limits of the prediction.

Suggested first pass: 1, 2, 3, 9. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 2; 3 course days later.*
Distance 16 m; displacement +2 m (right); final position 7 m.

02. Explain a worked example (worked)
Start 1 m, middle 7 m, range 4-10 m; constant velocity within the stated bounds.

03. Start plus change (model)
Displacement 8 m right; final position 11 m.

04. Predict leftward motion (practice)
6 m left; 12-6=6 m.

05. Match the time units (reason)
2×60=120 m. Must convert minutes to seconds before multiplying.

06. Read the range (reason)
Middle 10 m; lower 8 m; upper 12 m; half-width 2 m; full width 4 m.

07. A slow and a fast case (reason)
Lower 1×4=4 m; upper 3×4=12 m. Range 4-12 m under the stated constant-speed bounds.

08. Choose a method (mixed) *Review: Unit 1, Day 4; 1 course days later.*
Displacement -15 m. Second arrow points southeast with component ratio 7: 2.

09. Exit check (exit)
Central 8 m; range 7-9 m. Assumes steady velocity for the interval and the stated starting position/model.

10. STRETCH: Apply it another way (stretch)
Examples x_i=0 m, v=5 m/s and x_i=6 m, v=2 m/s; both endpoints 10 m.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 2; 3 course days later.*
Distance 17 m; displacement -3 m; final position +3 m.

02. Explain a worked example (worked)
Start 1 m, middle 7 m, range 4-10 m; constant velocity within the stated bounds.

03. Units and a starting position (model)
t=120 s; displacement 2400 m=2.40 km; final position 7.40 km.

04. Independent bounds (practice)
Central 20 m; half-width 1+5(0.2)=2 m; interval 18-22 m; full width 4 m.

05. A negative velocity (reason)
Lower 28+(-3.1)(10)=-3 m; upper 32+(-2.9)(10)=3 m; central 0 m, half-width 3 m.

06. Show how the bound grows (reason)
Central 4+2 t; lower 3+1.5 t; upper 5+2.5 t. At 4 s: central 12 m; range 9-15 m.

07. Input uncertainty or model error? (reason)
Gravity changes velocity; Earth moves; 3 D geometry and other omitted effects matter. Narrow input bounds do not remove model error.

08. Choose a method (mixed) *Review: Unit 1, Day 4; 1 course days later.*
Displacement 9 m; distance 21 m. An angle needs a reference direction/axis and rotation sense.

09. Exit check (exit)
Central 25 m; half-width 0.5+10(0.1)=1.5 m; interval 23.5-26.5 m; full width 3 m. Constant velocity and independent bounded inputs.

10. STRETCH: Apply it another way (stretch)
Central 14 m; lower 1.5+2.8×4=12.7 m; upper 2.5+3.2×4=15.3 m. These are bounds, not probabilities.